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A Success Story in Utilizing Big Data in Reporting Tier 3 Process Safety Incidents and Future Forward in Utilizing this Tool with AI to Save +10 Million Dollar Per Annum

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Abstract

Objectives/Scope: This paper introduces a novel approach to increased tier-3 incident reporting by tenfold.

Methods, Procedures, Process: This paper presents a transformative approach to process safety management through the integration of big data analytics and predictive modeling, with a focus on Tier 3 incident detection. The study outlines the development and deployment of a digital tool, capable of real-time monitoring and automated identification of safety system demands, particularly on Pressure Safety Valves (PSVs) and Safety Instrumented Functions (SIFs). The tool uncovered significant underreporting of Tier 3 incidents, revealing latent risks and enabling targeted maintenance interventions aligned with API RP 576 standards.

The developed tool is equipped with process parameter tags and their corresponding safe operating limits (SOLs) to detect any exceedance, demand on safety systems, or relief devices. This paper outlines the methodology for implementing this big data detection system, as applied in a real-world case study. It highlights the challenges encountered during deployment and the valuable results achieved in enhancing plant safety and operational reliability.

Results, Observations, Conclusions: Prior to the implementation of this tool, less than 10% of Tier-3 process safety incidents were identified despite field operator rounds, operational monitoring, and trend analysis. This left 90% of incidents undetected, many of which recurred and posed a high risk of escalating into Tier-1 incidents. The future direction involves integrating Artificial Intelligence (AI) to proactively analyze process parameters, identify potential activations of critical Emergency Shutdown (ESD) loops, and recommend operator actions. Operators will retain control by modifying these recommendations to ensure smooth operations. This predictive approach is expected to address 30% to 60% of plant trip scenarios, potentially saving over \$10 million annually.

Novel/Additive Information: This research presents a significant advancement in the detection and reporting of Tier-3 process safety incidents. The novel tool introduced in this study has successfully increased incident reporting by tenfold, addressing a critical gap where 90% of incidents previously went

undetected. Unlike other research that primarily focuses on preventing spurious trips, this study emphasizes a proactive approach by utilizing AI to prevent demands on safety systems. The proactive analysis of process parameters and identification of potential activations of critical ESD loops represent a forward-thinking strategy that significantly reduces the risk of plant trips. This innovative approach is expected to address 30% to 60% of plant trip scenarios, potentially saving over \$10 million annually.

Introduction

The oil and gas industry is inherently hazardous, necessitating a critical role for Process Safety in safeguarding people, assets, and the environment. This sector is characterized by multiple Major Accident Hazards (MAHs), primarily due to the flammable nature of hydrocarbons, which can lead to fires and explosions. Additionally, the presence of toxic materials such as hydrogen sulfide poses serious health risks. The industry also operates under extreme operating conditions such as those in low-density polyethylene production where the operating pressures is exceeding 2,500 barg. Loss of primary containment (LOPC) under such extreme pressure can lead to catastrophic consequences. These factors collectively contribute to a high-risk environment where even minor failures can escalate into catastrophic events.

Process Safety is fundamentally aimed at preventing MAHs that could result in loss of life, environmental harm, and substantial financial consequences. Traditionally, Process safety performance has been assessed using lagging indicators, which reflect incidents that have already occurred. While these indicators are essential for post-incident analysis and regulatory reporting, they offer limited value in preventing future occurrences.

Recognizing this limitation, the American Petroleum Institute (API) introduced the API Recommended Practice 754, which provides a structured framework for categorizing process safety indicators into four tiers. These tiers range from Tier 1 (major process safety events) to Tier 4 (more related to management system). The framework emphasizes the importance of leading indicators, which are proactive metrics that signal potential hazards before they escalate.

The figure below illustrates the customized structure of process safety indicators, highlighting the leading and lagging indicators based on API 754.



Figure 1—Company Leading and Lagging indicators based on API754 Framework

A strategic initiative was undertaken to enhance the detection and reporting of Tier 3 process safety incidents. Typically tier 3 is more concerned with the challenge to the protection and safety system. These tier 3 requires investigation to figure out the causes of challenge to safety system and prevent them in future. Consequently, this reduction in lagging indicators contributes to a corresponding improvement in leading indicators, reinforcing a safer and more resilient operational environment.

Leveraging advanced big data analytics, Company successfully developed a digital tool capable of real-time monitoring and automated detection of Tier 3 process safety events, with a particular focus on demands on Pressure Safety Valves (PSVs) and Safety Instrumented Functions (SIFs). In the absence of such a system, most of these events often go undetected, resulting in missed opportunities for investigation and root cause analysis. Consequently, latent design flaws or operational inefficiencies remain unresolved, posing long-term risks to plant safety and reliability. The implementation of big data methodologies has significantly enhanced the detection capabilities, enabling timely identification, reporting, and corrective action for Tier 3 incidents — thereby strengthening the overall process safety framework.

Literature Review

The integration of Big Data (BD) and Artificial Intelligence (AI) into process safety management has gained significant attention in recent years, particularly within the chemical and petrochemical industries. These technologies offer transformative potential in enhancing operational efficiency, predictive maintenance, and risk mitigation.

"Big Data" in Process Safety

"Big Data" refers to the vast volume of structured and unstructured data generated by industrial systems, including sensors, control systems, and enterprise applications. In the context of process safety, BD enables:

- Real-time monitoring of process parameters
- Historical trend analysis for incident investigation
- Predictive modeling for equipment reliability
- Automated anomaly detection

According to [Fornasiero et al. \(2021\)](#), while BD has been widely adopted in discrete manufacturing, its application in the process of industry includes refining, and petrochemical remaining limited but growing. The study highlights that BD is a prerequisite for AI, enabling the storage, access, and analysis of large datasets that traditional systems cannot handle efficiently.

Artificial Intelligence in Process Safety

AI technologies, including machine learning (ML), deep learning, and rule-based systems, are increasingly being explored for their ability to enhance Process Safety Management (PSM). Applications include:

- Process Hazard Analysis (PHA) revalidation
- Dynamic risk assessment
- Predictive maintenance
- Alarm management and prioritization

[Balasubramanian et al. \(2024\)](#) emphasize that AI can significantly improve the efficiency of PSM studies, especially when integrated with digitalization initiatives. However, challenges such as data labeling, model training, and system integration must be addressed for successful implementation.

Industry 4.0 and Digital Transformation

The broader context of Industry 4.0—which includes AI, BD, Internet of Things IoT, and cyber-physical systems—has accelerated the digital transformation of the chemical industry. [Laska and Karwala \(2023\)](#) discuss both the opportunities and risks associated with AI adoption, including ethical concerns, cybersecurity, and workforce adaptation. They argue that while AI can enhance safety and efficiency, it must be implemented with robust governance frameworks.

Gaps and Opportunities

Despite the promising potential, several gaps remain:

- Limited deployment of AI/BD in core Process safety areas such as relief device activation and control system
- Lack of standardized frameworks for AI integration in PSM
- Need for interdisciplinary collaboration between data scientists and process engineers
- Underutilization of historical data for predictive analytics

This paper addresses some of these gaps by presenting a real-world case study from one of the companies, demonstrating how BD and AI can be effectively leveraged to enhance Tier 3 incident detection and prevention.

Methodology

System Architecture and Data Integration

In the context of this project, "big data" refers to the comprehensive set of all of the instrumentation readings connected to the Distributed Control System (DCS) and by the way to historian software such as the PI system. These include, but are not limited to, pressure readings, flow rates, temperature measurements, valve positions, level reading, and control signals from Emergency Shutdown (ESD) systems, and others.

The developed digital tool is seamlessly integrated with Company's Process Information (PI) system, forming the backbone of a robust, data-driven process safety monitoring framework. This integration enables the system to continuously track a wide range of critical process parameters and compare them against predefined Safe Operating Limits (SOLs). Any deviation from these limits is automatically flagged as a potential Tier 3 process safety incident. This exceedance is such as an activation of a pressure relief device, or a demand on a safety system or other deviation. This real-time detection capability ensures that early warning signs are not overlooked, allowing for timely investigation and corrective action.

The system architecture is designed to support both operational responsiveness and analytical depth, and includes the following key components:

- Real-time data acquisition from strategically placed field instruments, including pressure transmitters, flow meters, and control valves, ensuring high-resolution visibility into process dynamics.
- Historical data analysis capabilities that allow for trend validation, incident reconstruction, and root cause analysis, thereby supporting both proactive and retrospective safety management.
- Predictive modeling algorithms for PSVs that lack direct instrumentation, enabling accurate estimation of pressure conditions using hydraulic calculations and surrogate data sources.
- Manual validation workflows that empower operations personnel to review flagged events, verify anomalies through trend analysis, and confirm incident reports before escalation. This human-in-the-loop approach ensures data integrity and operational accountability.

Together, these components form a comprehensive digital safety infrastructure that enhances situational awareness, supports compliance with industry standards, and enables a shift from reactive to predictive process safety management.

Predictive Modeling for PSV Monitoring

For PSVs without direct pressure transmitters, a predictive model was developed using hydraulic principles. The formula used is:

$$P_{PSV} = K Q^2 + P_{PT} + \Delta H \cdot \rho \cdot g$$

Where:

- P_{PSV} : Estimated pressure at the PSV
- K : Constant derived from hydraulic and piping configuration
- Q : Volumetric flow rate from the flowmeter
- P_{PT} : Pressure reading from the nearest pressure transmitter
- $\Delta H \cdot \rho \cdot g$: Static pressure loss due to elevation difference

This formula enabled reasonably accurate estimation of the pressure at PSVs without direct instrumentation, ensuring comprehensive coverage of relief system monitoring. The methodology was applied in a real-world case study at some of the units, where challenges such as data integration, instrumentation gaps, and calibration were addressed. The implementation yielded valuable insights and significantly improved the detection and reporting of Tier 3 incidents.

The Big Data format includes multiple inputs especially for the causes were Pressure transmitter (PT) is not measuring the same value on PSV. This requires data validation tools.

Table 1—template of Big Data set for PSVs

PSV Tag Number	P&ID No.	Max Operating Pressure	Set Pressure	PT MAX Value	Pressure at PSV	Number of Demands
XX-PSV-0001	XXXXXXX	7.5	9	8	9.2	2

By applying this model, Company ensured comprehensive coverage of all PSVs, eliminating blind spots in the relief system monitoring.

To ensure the reliability of this model, a validation was conducted comparing model-predicted values against actual measurements from PSVs equipped with direct pressure instrumentation. The results demonstrated an average error rate of $\pm 3\%$, which falls well within acceptable engineering tolerances for relief system monitoring. Notably, this aligns with API Recommended Practice, which even permits up to a 3% deviation in PSV set pressure accuracy.

This model for sure is most feasible solution for many equipment as it is not applicable to install pressure measurement connecting DCS to all of the PSV within the plant. Honestly even after this model, still there are some of PSVs that cannot measure directly or indirectly because it has local measurement. The error has multiple factors mainly due to instrumentation errors themselves. As an example, Coriolis mass flowmeters has error less than 0.1% while orifice has between 1% to 2 %. So, if the flowmeter available is orifice, the error will be higher.

Data Validation and Incident Confirmation

The data validation process within the developed tool is structured into two distinct stages to ensure the accuracy, reliability, and integrity of safety-critical information.

Stage 1: Instrumentation Validation, this initial phase involves the cross-verification of readings from multiple instrumentation sources. For example, pressure readings from an Emergency Shutdown (ESD) transmitter are validated against corresponding values from the Distributed Control System (DCS) transmitter. This redundancy helps identify discrepancies caused by sensor drift, calibration errors, or signal noise. In addition to direct comparison, this stage incorporates engineering calculations to assess consistency and plausibility of the measured values, thereby reinforcing the robustness of the data acquisition process.

Stage 2: Manual Validation and Operator Confirmation, given the complexity and dynamic nature of process operations, automated detection alone may not capture the full context of an event. Therefore, a manual validation layer is incorporated, wherein experienced operators review historical trends, analyze process behavior, and confirm anomalies before incidents are formally reported. This step ensures that only genuine Tier 3 process safety events are escalated for investigation, minimizing false positives and enhancing the credibility of the reporting system.

Results and Discussion

Uncovering Hidden Risks and enhancing PSV maintenance

The deployment of the big data-driven tool at Company revealed a substantial gap in the historical reporting of Tier 3 process safety incidents. Prior to implementation, less than 10% of Pressure Safety Valve (PSV) activations were identified through conventional manual monitoring methods such as operator rounds and trend reviews. This limited visibility meant that the vast majority of safety system demands went undocumented, leaving critical events uninvestigated and unresolved.

Through comprehensive data analysis, the tool uncovered that certain PSVs had been activated multiple times per week over a span of more than 15 years—without any formal reporting, inspection, or maintenance follow-up. This chronic underreporting posed a significant latent risk, as repeated activations of relief devices can indicate underlying design flaws, control loop instability, or operational inefficiencies. Left unaddressed, these issues have the potential to escalate into Tier 1 incidents with severe consequences.

The insights generated by the tool not only highlighted the limitations of traditional monitoring practices but also underscored the value of automated, data-driven detection in strengthening process safety. By enabling real-time visibility and retrospective analysis, the system has transformed incident reporting from a reactive to a proactive discipline, allowing for timely intervention, root cause analysis, and continuous improvement.

The implementation of the big data tool at the company provided critical insights into the operational reliability of Pressure Safety Valves (PSVs), revealing previously overlooked patterns of failure and maintenance gaps. Historically, several PSVs that failed bench tests were attributed to procedural mishandling or isolated anomalies during testing. However, the tool's continuous monitoring and historical data analysis uncovered a more systemic issue: repeated activations of certain PSVs over extended periods without any corresponding inspection, recalibration, or maintenance intervention.

This finding was noncomplying with the guidelines outlined in API Recommended Practice 576, which stipulate that PSVs—particularly those operating in dirty or fouling service—must undergo at least a visual inspection of internal components following each activation. Failure to adhere to these inspection protocols can compromise the integrity of the relief system, increasing the risk of malfunction in case of overpressure.

Following the deployment of the tool, Company was able to identify and rectify critical gaps. Historical analysis revealed that over 90% of PSV activations had gone undetected, primarily involving a small group of high-frequency "bad actor" PSVs. Four PSVs were found to be demanding multiple times per months, yet had never been formally reported or investigated. This chronic underreporting posed significant risks to

plant safety and reliability. Additionally, the tool uncovered that the number of previously undiscovered PSV pop-ups—Excluding the repeated times for the same PSV—was higher than the number of PSVs captured by traditional methods, highlighting the limitations of manual monitoring. It worth highlighting that the full judgment for bench test enhancement will be evaluated by the next turnaround

Historical Data Utilization for Risk Forecasting

One of the most impactful capabilities of the developed tool lies in its ability to harness and analyze extensive historical process data to forecast future risks with a high degree of accuracy. This predictive functionality is particularly valuable in scenarios involving deferred maintenance or postponed shutdowns, where equipment may operate in a downgraded state for extended periods.

For instance, if a Pressure Safety Valve (PSV) is scheduled for inspection but its maintenance is deferred due to operational constraints, the tool can retrospectively calculate the PSV's demand rate over the past five years using archived data from the PI system. By evaluating the frequency and conditions of past activations, the system can estimate the probability of future demands during the deferment period. This predictive insight enables engineers and safety professionals to perform quantitative risk assessments grounded in empirical data rather than assumptions.

Such analysis supports informed decision-making regarding whether to proceed with the deferral, implement temporary safeguards, or prioritize the equipment for early intervention. It also enhances compliance with internal risk management protocols and external regulatory standards by providing a defensible, data-backed rationale for operational decisions. Ultimately, this capability transforms historical data from a passive record into an active tool for risk mitigation and strategic planning.

Design Deficiencies and Control Loop Optimization

The tool also identified design deficiencies contributing to PSV activations. Many incidents were traced to control loop inefficiencies, particularly at high-pressure/low-pressure interfaces. These findings prompted:

- Detailed incident investigations
- Management of Change proposals
- Updates to Standard Operating Procedures (SOPs) as a mitigation action till the remedial action is implemented
- Temporary mitigation measures pending design upgrades

This proactive approach ensures that systemic issues are addressed before they lead to major incidents.

Start up troubleshooting

A significant proportion of demands on PSVs and safety systems were observed by this tool to occur during transitional operational states, such as startup, shutdown, and cyclic operations. These phases are inherently complex and dynamic, often requiring operators to manage multiple simultaneous tasks under time-sensitive conditions. As a result, the likelihood of overlooking subtle process deviations or early signs of equipment stress increases substantially.

According to the Center for Chemical Process Safety (CCPS), nearly 50% of process safety incidents are reported to occur during start up and shut down. This statistic underscores the elevated risk profile associated with non-steady-state operations and highlights the need for enhanced monitoring capabilities during such times.

The deployment of Big Data tool addresses this challenge by providing continuous, high-resolution monitoring of process parameters, even during operational transitions. The system's ability to detect anomalies in real time — regardless of operator workload — represents a critical enhancement to traditional safety practices. By reducing reliance on human vigilance and enabling automated detection of deviations,

the tool significantly improves safety outcomes and supports more effective incident prevention during the most vulnerable phases of plant operation.

Safety Instrumented Function (SIF) Validation

The tool also plays a pivotal role in validating Safety Instrumented Function (SIF) demand rates, which are traditionally difficult to extract and quantify from event logs due to the presence of signal noise, spurious trips, and inconsistent data formatting. These challenges often obscure the true frequency and nature of safety system activations, making it difficult to assess performance and compliance accurately.

To overcome these limitations, the system employs a multi-source data validation approach, cross-referencing readings from various transmitters and control points. By comparing values from independent instrumentation channels—such as process transmitters, DCS logs, and ESD systems—the tool effectively filters out false positives and isolates genuine demands on the SIFs. This ensures that only valid activations are recorded and analyzed.

The resulting high-fidelity demand statistics support several critical safety and compliance functions:

- Safety Integrity Level (SIL) verification: Accurate demand data enables periodic reassessment of SIL ratings, ensuring that the low demand SIF are not changed to high demand.
- Compliance with IEC 61511 standards: The tool's validation capabilities align with the requirements of IEC 61511 for functional safety in the process industry, supporting audit readiness and regulatory reporting.
- Lifecycle management of safety systems

By enhancing the accuracy and reliability of SIF demand data, the tool strengthens the foundation of Company's safety instrumented system strategy and reinforces its commitment to operational excellence and regulatory compliance.

Compliance with standard Benefits

The enhanced visibility into Tier 3 incidents also supports compliance with standards such as:

- API RP 754: Process Safety Performance Indicators
- API RP 576: Inspection of Pressure-Relieving Devices
- IEC 61511: Functional Safety for the Process Industry Sector

By automating data collection and reporting, the tool reduces the administrative burden on safety teams and ensures timely submission of compliance documentation.

Future Work: AI Integration for Predictive Safety

Real-Time Prediction and Operator Support

The next phase involves integrating Artificial Intelligence (AI) to transition from reactive detection to predictive prevention. The AI module will analyze process parameters in real time, identify patterns that precede safety system activations, and recommend corrective actions.

For example, in a compressor knock-out drum, a sudden increase in level due to upstream disturbances will trigger high level alarm. However, the alarm does not differentiate between slow and rapid increases. With AI integration, the system can:

- Detect the rate of change
- Assess urgency

- Recommend / take actions such as adjusting control valves or starting backup pumps

Operators will receive confirmation message for AI-recommended actions, maintaining human oversight while benefiting from machine precision. As an example, to confirm opening LIC on compressor KOD from 40% to 80% or to start a specific pump. Operators will retain the flexibility to adjust these recommendations to maintain operational continuity.

Integration with Advanced Process Control (APC)

The AI module will be integrated with Advanced Process Control (APC) systems to enable closed-loop responses. This includes:

- Dynamic adjustment of control parameters
- Automated execution of mitigation strategies
- Feedback loops for continuous learning

This integration will enhance process stability, reduce manual intervention, and improve response times during critical events.

Economic Impact and Strategic Value

Preliminary models suggest that AI-driven interventions could prevent 30% to 60% of plant trips, resulting in:

- Reduced the demand on safety system which reduces the process safety incidents
- Reduced downtime and enhanced plant availability
- Enhanced asset integrity

The estimated annual savings exceed \$10 million, making this initiative not only a safety enhancement but also a strategic investment.

Human-Machine Collaboration

One of the key enablers of success in this initiative has been the collaborative interface between human operators and digital tools. While the system automates detection and analysis, final validation and decision-making remain with experienced personnel. This human-in-the-loop approach ensures that:

- Operators retain control over safety-critical decisions
- AI recommendations are contextualized with operational knowledge
- Trust in the system is gradually built through transparency and accuracy

This model aligns with best practices in sociotechnical system design, where technology augments rather than replaces human expertise.

To ensure successful AI integration into process safety systems, a dual focus was placed on workforce engagement and structured deployment.

Workforce Engagement

The success of AI integration in process safety relies heavily on workforce readiness, trust, and continuous engagement. To support this, a structured change management program has been implemented to ensure smooth adoption and long-term sustainability. This initiative is being managed as a dedicated program,

receiving focused attention from the company. It includes comprehensive online training modules, open sessions across all company functions to gather feedback, and an online portal that provides ongoing support and resources for AI-related operations.

The implementation roadmap includes:

- Short-Term (Q1-Q2 2026): Pilot deployment on compressor KODs with a target KPI of <10% false alarms.
- Mid-Term (Q3-Q4 2026): Integration with APC systems and the target KPI >90% prediction accuracy.
- Long-Term (Q4 2026 onward): Full rollout across the plant, aiming for 30-60% trip reduction and >\$10M in annual savings.

Conclusion

This experience presents a transformative approach to enhancing process safety performance through the integration of big data analytics and predictive modeling. By equipping the system with process parameter tags and operational limits, the tool successfully identified previously undetected Tier 3 process safety incidents, increasing detection rates by tenfold. This breakthrough addressed a critical gap in traditional monitoring methods, where over 90% of PSV demands had gone unnoticed for years, posing significant escalation risks.

The tool not only improved incident visibility but also uncovered reliability issues in PSVs, corrected misdiagnosed root causes, and aligned inspection practices with API 576 standards. Furthermore, it revealed design deficiencies in control loops, particularly during transitional operations such as startup and shutdown, leading to targeted Management of Change actions and SOP updates.

Looking ahead, the integration of AI represents the next frontier. By proactively analyzing process parameters and predicting potential activations of Emergency Shutdown (ESD) loops, the system will empower operators with actionable insights to prevent plant trips. This forward-thinking strategy is expected to mitigate 30% to 60% of trip scenarios, with an estimated annual savings exceeding \$10 million.

Novel Contribution

This study contributes to the growing body of knowledge on digital transformation in process safety by:

- Demonstrating a scalable model for Tier 3 incident detection using big data
- Introducing predictive modeling for PSVs without direct instrumentation
- Validating the use of historical data for risk forecasting
- Proposing a roadmap for AI integration in safety-critical operations
- Highlighting the importance of human-machine collaboration and ethical AI

By addressing both technical and organizational dimensions, this work sets a new benchmark for data-driven safety management in the chemical and petrochemical sectors.

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